Models with vorticity

A diffusive exclusion process with vorticity 1

Given a configuration η and $\{x, y\} \in \mathcal{E}_N$

$$\eta^{x,y}(z) := \begin{cases} \eta(y) & \text{if } z = x\,, \\ \eta(x) & \text{if } z = y\,, \\ \eta(z) & \text{otherwise}\,. \end{cases}$$

The generator of the dynamics is given on $f : \{0,1\}^{\Lambda_N} \to \mathbb{R}$ by

$$\mathcal{L}_N f(\eta) = \sum_{(x,y) \in E_N} c_{x,y}(\eta)\left[f(\eta^{x,y}) - f(\eta)\right]. \tag{1}$$

$$c_{x,y}(\eta) = c_{x+z,y+z}(\tau_z \eta)\,. \tag{2}$$

A diffusive exclusion process with vorticity 2

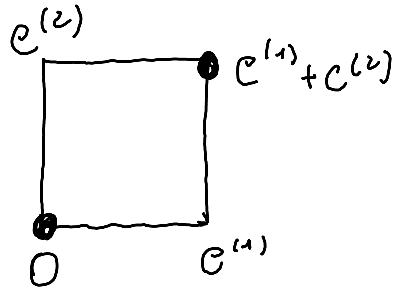
The rates are

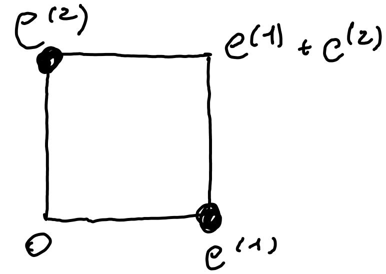
$$c_{x,y}(\eta) := \eta(x)\big(1-\eta(y)\big) + \eta(x)\left[\tau_{\mathfrak{f}^+(x,y)}g(\eta) - \tau_{\mathfrak{f}^-(x,y)}g(\eta)\right] , \qquad (3)$$

where the function g is a local function depending just on the occupation numbers at the vertices of the face $\{0, e^{(1)}, e^{(1)}+e^{(2)}, e^{(2)}\}$. More precisely, we have

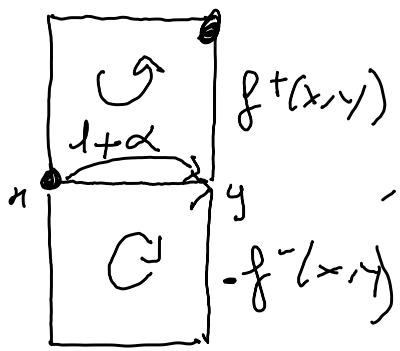
$$g(\eta) = \begin{cases} \alpha & \text{if } \eta(0)=\eta(e^{(1)}+e^{(2)})=1 \text{ and } \eta(e^{(1)})=\eta(e^{(2)})=0\,, \\ \alpha & \text{if } \eta(e^{(1)})=\eta(e^{(2)})=1 \text{ and } \eta(0)=\eta(e^{(1)}+e^{(2)})=0\,, \\ 0 & \text{otherwise}\,, \end{cases} \qquad (4)$$

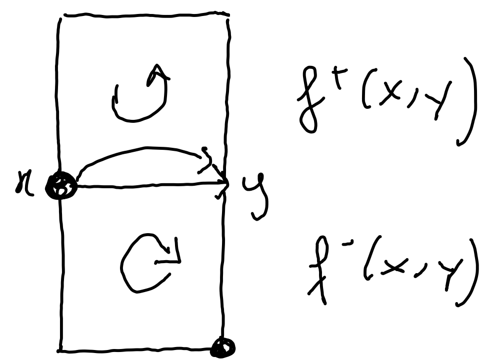
where α is a real parameter such that $|\alpha| < 1$.

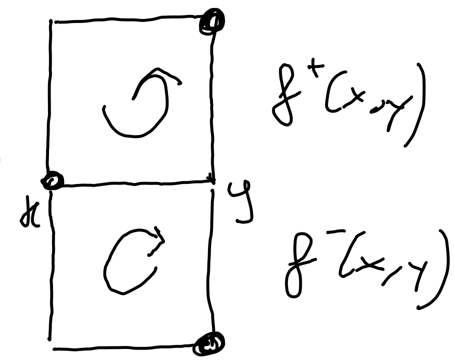
$g(\eta)=\alpha$ 

, $g(\eta)=\alpha$ 

, 0 otherwise.

$1+\alpha$ 

, $1-\alpha$ 

, 1 

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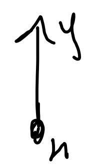
When a particle rotate along anticlockwise face the rate is increased by $+\alpha$, viceversa by $-\alpha$.

Along the direction $e^{(2)}$ 

it is the same.

Model interpretation 1

Particles perform a simple exclusion process, but the faces containing exactly 2 particles located at sites which are not nearest neighbors let the particles rotate anticlockwise with rate $\alpha > 0$ and clockwise with rate $-\alpha < 0$.

The model is invariant under rotation.

Model interpretation 2

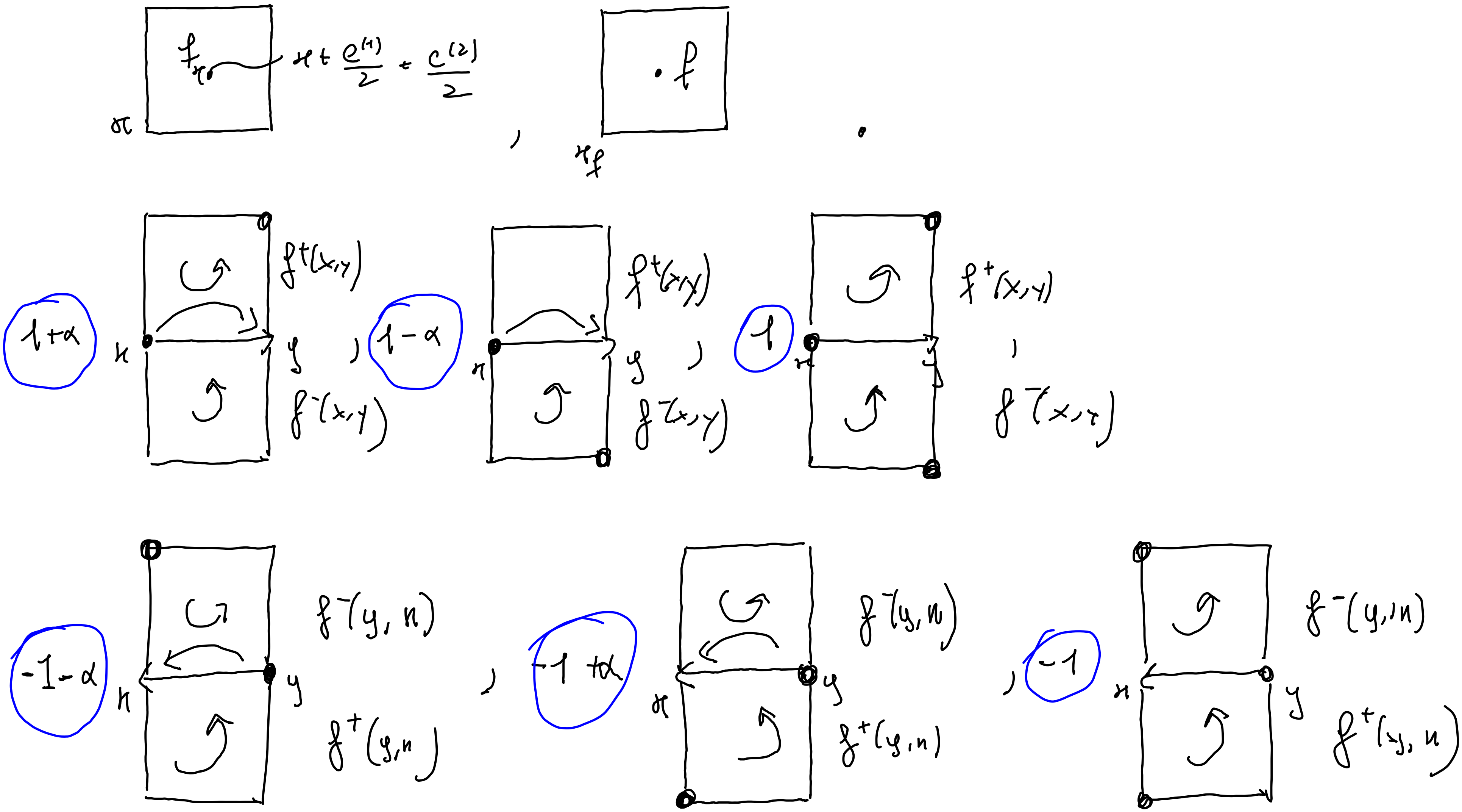
The instantaneous current is

$$j_\eta(x,y) = [\eta(x) - \eta(y)] + [\tau_{\mathfrak{f}^+(x,y)} g(\eta) - \tau_{\mathfrak{f}^-(x,y)} g(\eta)], \tag{5}$$

where $g(\eta)$ is also $\alpha\eta(0)(1-\eta(e^{(1)}))\eta(e^{(1)}+e^{(2)})(1-\eta(e^{(2)})) + \alpha(1-\eta(0))\eta(e^{(1)})(1-\eta(e^{(1)}+e^{(2)}))\eta(e^{(2)})$.

Remark: $\operatorname{div} j_\eta(x) = \operatorname{div} j^\nabla_\eta(x)$, the parts of g is divergence free.

Reminder: $\tau_{\mathfrak{f}} := \tau_x$ with $\mathfrak{f}$ centered in $x + \frac{e_1}{2} + \frac{e_2}{2}$.



Observations: ∘) $f^{\pm}(x,y) = f^{\mp}(y,x)$.

∘) the current $j_\eta(x\,y) = \sum_{f^+(x,y)} g(\eta) - \sum_{f^-(x,y)} g(\eta)$ contributes with $\pm\alpha$ when particles rotate anticlockwise/clockwise in the same way in both directions.

Invariant measure 1

Bernoulli $\mu_N^\rho(\eta) = \prod_{x\in V_N} \rho^{\eta(x)}(1-\rho)^{(1-\eta(x))}$ is invariant but not reversible.

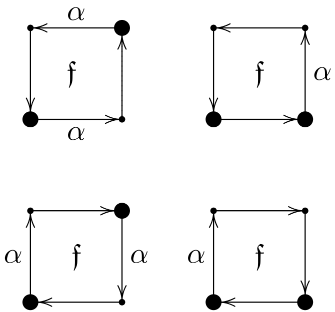
We need to verify $\displaystyle\sum_{(x,y)\in E_N} c_{x,y}(\eta) = \sum_{(x,y)\in E_N} c_{y,x}(\eta^{x,y})\,,$ reduced to

$$\sum_{\mathfrak{f}\in\mathcal{F}_N}\sum_{(x,y)\in f^{\circlearrowleft}} \eta(x)\left[\tau_{\mathfrak{f}}g(\eta)+\tau_{\mathfrak{f}}g(\eta^{x,y})\right] = \sum_{\mathfrak{f}\in\mathcal{F}_N}\sum_{(x,y)\in f^{\circlearrowright}} \eta(x)\left[\tau_{\mathfrak{f}}g(\eta)+\tau_{\mathfrak{f}}g(\eta^{x,y})\right]\,, \tag{6}$$

Invariant measure 2

It is enough to verify for a face

$$\sum_{(x,y)\in f^{\circlearrowleft}} \eta(x)\left[\tau_{\mathfrak{f}} g(\eta) + \tau_{\mathfrak{f}} g(\eta^{x,y})\right] = \sum_{(x,y)\in f^{\circlearrowright}} \eta(x)\left[\tau_{\mathfrak{f}} g(\eta) + \tau_{\mathfrak{f}} g(\eta^{x,y})\right] ,$$



The empirical measure $\pi^N(\eta, du)$ is defined by

$$\pi^N(\eta, du) := \frac{1}{N^d} \sum_{x \in V_N} \eta(x)\delta_x(du)\,, \tag{7}$$

with an initial sequence of probability measures μ_N on Σ_N associated to the density profile ρ if for any $f \in C(\Lambda)$

$$\lim_{N\to+\infty} \mu_N \left(\eta \in \Sigma_N : \left| \int_{\mathbb{T}^2} f(u)\, d\pi^N(\eta, du) - \int_{\mathbb{T}^2} f(u)\rho(u)du \right| > \varepsilon \right) = 0, . \tag{8}$$

$\forall \varepsilon > 0$, i.e. the sequence of measures $\pi^N(\eta, du) \in \mathcal{M}^+(\mathbb{T}^2)$ converges weakly in probability with respect to μ_N to the measure $\rho(u)du \in \mathcal{M}^+(\mathbb{T}^2)$.

Hydrodynamics for the empirical measure 2

Theorem

Let η_t be the Markov process with generator given by (1) *with rates given in* (3) *multiplied by a factor of N^2. Suppose to start the process from a sequence μ_N of probability measures which are associated (according to* (8)*) to a density profile $\rho^* : \mathbb{T}^2 \to [0,1]$. Then, for any $f \in C(\mathbb{T}^2)$ and any $\varepsilon > 0$, it holds*

$$\lim_{N\to+\infty} \mathbb{P}_{\mu_N}\left(\eta \in \mathcal{D}([0,T],\Sigma_N) : \left|\int_{\mathbb{T}^2} f(u)\, d\pi_N(\eta_t, du) - \int_{\mathbb{T}^2} f(u)\rho_t(u)du\right| > \varepsilon\right) = 0\,, \tag{9}$$

where $\rho_t(u)$ is the unique weak solution of the Cauchy problem

$$\begin{cases} \partial_t\rho_t(u) = \Delta\rho_t(u)\,, \forall u \in \mathbb{T}^2, \forall t > 0, \\ \rho_0(u) = \rho^*(u)\,, \forall u \in \mathbb{T}^2. \end{cases} \tag{10}$$

Proof Hydrodynamics for the empirical measure

The proof is essentially is essentially unchanged w.r.t. SSEP. Because:

$$\int_{\mathbb{T}^d} f \, d\pi_N^t(\eta) - \int_{\mathbb{T}^d} f \, d\pi_N^0(\eta) = -N^{2-d} \int_0^t ds \sum_{x \in V_N} \operatorname{div} j_{\eta_s}(x) \, f(x) \, ds + o(1) \, .$$

from $\operatorname{div} j_\eta(x) = \operatorname{div} j_\eta^\nabla(x)$ and with discrete integrations by part

$$\int_{\mathbb{T}^d} f \, d\pi_N^t(\eta) - \int_{\mathbb{T}^d} f \, d\pi_N^0(\eta) = N^{2-d} \int_0^t ds \sum_{x \in V_N} \tau_x \eta(0) \, \Delta_N f(x) \, ds + o(1) \, .$$

That is the same of SSEP.

Integrated current field

<u>*Reminders*</u>:

$\mathcal{N}_t(x,y)$:= number of particles that jumped from site x to site y up to time t. The *current flow* across the bond (x,y) is

$$J_t(x,y) := \mathcal{N}_t(x,y) - \mathcal{N}_t(y,x), \text{ note that } J_t(x,y) = -J_t(y,x). \tag{11}$$

Key observation:

$M_t(x,y) = J_t(x,y) - N^2 \int_0^t j_{\eta(s)}(x,y)ds$ is a martingale.

The *integrated current field* is the field acting on continuous vector fields $G : \mathbb{T}^2 \to \mathbb{R}^2$ in the following way, $G_N(x,y) := \int_{(x,y)} G \cdot dl$,

$$\mathcal{J}_t^N(G) := \frac{1}{N^d} \sum_{(x,y)\in E_N} G_N(x,y)\mathcal{N}^N_{x,y}(t) = \frac{1}{N^d} \sum_{(x,y)\in \mathcal{E}_N} G_N(x,y)J^N_{x,y}(t), \tag{12}$$

Current topology 1

Consider on $C^\infty(\mathbb{T}^2, \mathbb{R}^2)$ linear operator $\mathcal{L} = (1 - \Delta)$, its eigenvalues are

$$\mathcal{L} h_z = \gamma_z h_z, \text{ where } \gamma_z = 1 + 4\pi^2 |z|^2.$$

We consider the scalar product between two fields as

$$\langle \phi, \phi' \rangle_{-k} = \sum_{j=1}^{d} \sum_{z \in \mathbb{Z}^2} \gamma_z^{-k} \phi(I^{j,z}) \phi'(I^{j,z}), \tag{13}$$

$I^{j,z}$ is the vec. field with i-th component $I_i^{j,z} := \delta_{i,j} h_z$. Therefore, the space $\mathcal{H}^2_{-k}$ consists of all functionals such that

Dual of $\mathcal{H}^2_k = W^{k,2}$

$$\|\phi\|^2_{-k} = \sum_{j=1}^{2} \sum_{z \in \mathbb{Z}^2} \gamma_z^{-k} \phi^2(I^{j,z}) < \infty. \tag{14}$$

Note $\mathcal{H}^2_{-k}$ is also the completion of $\mathcal{H}^2_0$ w.r.t. $\langle \cdot, \cdot \rangle_{-k}$.

Current topology 2

We consider $\mathcal{J}_t^N$ in the Sobolev space $\mathcal{H}^2_{-k^*}$, k^* from tightness.

A trajectory $\{(\mathcal{J})_t^N\}_{t\in[0,T]}$ will be in the space of càdlàg trajectories $\mathcal{D}([0,T],\mathcal{H}^2_{-k^*})$.
$\mathbb{P}_N := \mathbb{P}_{\mu_N} \circ (\mathcal{J}^N)^{-1}$ the probability measure on $\mathcal{D}([0,T],\mathcal{H}^2_{-k^*})$ and by $\mathbb{E}_N$ the expectation with respect to $\mathbb{P}_N$.

Hydrodynamics for the integrated current field 1

$$\eta_t(x) - \eta_0(x) + \nabla \cdot J_t^N(x) = 0\,, \qquad \forall x \in V_N\,, t > 0\,,$$

discrete continuity equation relating the emp. meas. and the current field

$$\int_{\mathbb{T}^2} f d\pi^N(\eta_t) - \int_{\mathbb{T}^2} f d\pi^N(\eta_0) - \mathcal{J}_t^N(\nabla f) = 0\,, \qquad \forall f \in C^1(\mathbb{T}^2)\,, \forall t \in [0,T]\,. \tag{15}$$

Considering a test vector field G we obtain the martingales

$$M_G^N(t) = \mathcal{J}_t^N(G) - \frac{N^{2-d}}{2}\int_0^t \sum_{(x,y)\in E_N} j_{\eta_s}(x,y) G_N(x,y) ds, \tag{16}$$

by some discrete integration and the form of the instantaneous current

$$M_G^N(t) = \mathcal{J}_t^N(G) - N^{2-d}\int_0^t ds \left(\sum_{x\in V_N} \eta_s(x)\nabla\cdot G_N(x) + \sum_{\mathfrak{f}\in\mathcal{F}_N} \tau_{\mathfrak{f}} g(\eta_s) \sum_{(x,y)\in f^{\circlearrowleft}} G_N(x,y)\right). \tag{17}$$

Hydrodynamics for the integrated current field 2

From

$$\begin{cases} N^2 \nabla \cdot G_N(x) = \nabla \cdot G(x) + O(1/N^2)\,, \\ N^2 \sum_{(x,y)\in f^{\circlearrowleft}} G_N(x,y) = \nabla^{\perp} \cdot G(z) + O(1/N^2), \end{cases} \tag{18}$$

where $\nabla^{\perp} = (-\partial_y, \partial_x)$, and a replacement lemma

$$\frac{1}{N^2} \sum_{\mathfrak{f}\in\mathcal{F}_N} \tau_{\mathfrak{f}} g(\eta_s) \sum_{(x,y)\in f^{\circlearrowleft}} G_N(x,y) \approx \frac{1}{N^2} \sum_{x\in V_N} \mathbb{E}_{\rho(x/N,s)}[g(\eta_s)] \nabla^{\perp} \cdot G(z), \tag{19}$$

$\rho(u,t)$ is the traj. followed by $\pi_N^t(\eta, du)$ ($\pi_N^t(\eta, du) \approx \pi^t(du) = \rho(u,t)du$, $\rho(u,t)$ solves the heat eq.), we have (in probability) a weak form of

$$\mathcal{J}_t(G) = \int_0^t ds \int_{\mathbb{T}^2} du\ J(\rho(u,s)) \cdot G(u),\ \forall t \in [0,T],\ \text{with } J(\rho) = -\nabla\rho - \nabla^{\perp} a(\rho), \tag{20}$$

where $a(\rho(u,s)) = \mathbb{E}_{\rho(u,s)}[g(\eta)] = \alpha\rho^2(u,s)(1-\rho(u,s))^2$.

Hydrodynamics for the integrated current field 3

- First observe that

$$\partial_t \rho = \nabla \cdot (-J(\rho)) = \Delta \rho. \tag{21}$$

- Define the antisymmetric matrix $A(\rho) = \begin{pmatrix} 0 & -a'(\rho) \\ a'(\rho) & 0 \end{pmatrix}$, we can rewrite

$$J(\rho) = -\nabla \rho - A(\rho) \nabla \rho \,, \tag{22}$$

then we have a generalized Fick'law

$$J(\rho) = -\mathcal{D}(\rho) \nabla \rho, \;\; \mathcal{D}(\rho) = \begin{pmatrix} 1 & -a'(\rho) \\ a'(\rho) & 1 \end{pmatrix}. \tag{23}$$

Current Theorem

Theorem

Let η_t be the Markov process with generator given by (1) with rates given in (3) multiplied by N^2. Suppose to start the process from a sequence of probability measures μ_N which are associated (according to (8)) to a density profile $\rho^* : \mathbb{T}^2 \to [0,1]$. Then, for any C^1 vector field G on $\mathbb{T}^2$ and for any $\varepsilon > 0$, it holds

$$\lim_{N\to+\infty} \mathbb{P}_{\mu_N}\left(\eta \in \mathcal{D}([0,T],\Sigma_N) : \left|\mathcal{J}_t^N(G) - \int_{\mathbb{T}^2} du \int_0^t ds\, J(\rho_s(u)) \cdot G(u)\right| > \epsilon\right) = 0\,, \forall t \in [0,T], \tag{24}$$

where

$$J(\rho) = -\nabla\rho - A(\rho)\nabla\rho\,, \tag{25}$$

and $\rho_t(u)$ is the unique weak solution of the Cauchy problem (10).

Proof: tightness of $\{\mathbb{P}_N\}_N$ + characterization of limits points.

Proof: tightness

Skorohod top. w.r.t. unif. modulus of continuity $w_\delta(\mathcal{J})$ on $\mathcal{D}([0,T],\mathcal{H}^2_{-k})$

$$w_\delta(\mathcal{J}) = \sup_{\substack{|t-s|\leq\delta \\ 0\leq s,t\leq T}} \|\mathcal{J}_t - \mathcal{J}_s\|_{-k}. \tag{26}$$

Theorem

A sequence of probability measures $\{\mathbb{P}_N, N\geq 1\}$ defined on $\mathcal{D}([0,T],\mathcal{H}^2_{-k})$ is tight if, for every $0\leq t\leq T$ and for every $\varepsilon>0$, we have

1. $\lim\limits_{\ell\to\infty}\limsup\limits_{N\to\infty}\mathbb{P}_N\left(\sup\limits_{0\leq t\leq T}\|\mathcal{J}_t\|_{-k}>\ell\right)=0;$
2. $\lim\limits_{\delta\to 0}\limsup\limits_{N\to\infty}\mathbb{P}_N\left(w_\delta(\mathcal{J})\geq\varepsilon\right)=0 \qquad \forall\varepsilon>0.$

Proof: characterization of the limit points 1

Theorem

Let $\mathbb{P}$ be a limit point of the sequence $\{\mathbb{P}_N\}_N$. Then, for $k > k^*$,

$$\mathbb{P}\left(\mathcal{J}_t \in \mathcal{C}([0,T], \mathcal{H}^2_{-k}) : F(G,t,\rho^*) = 0, \quad \forall t \in [0,T], \forall\, G \in C^1(\mathbb{T}^2)\right) = 1, \tag{27}$$

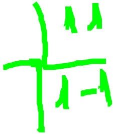
where

$$F(G,t,\rho^*) = \mathcal{J}_t(G) - \int_0^t ds \int_{\mathbb{T}^2} du \; \left(\rho_s(u)\nabla \cdot G(u) + 2\alpha\rho_s^2(u)(1-\rho_s(u))^2\nabla^\perp \cdot G(u)\right), \tag{28}$$

with $\rho_t(u)$ solving the Cauchy problem (10) because of Theorem 1.

Proof: characterization of the limit points 2

$\ell \in \mathbb{N}$, $B^{\ell}_{pe_1,qe_2}(x)$ is the box $[x_1, x_1 + \ell p e_1] \times [x_2, x_2 + \ell q e_2]$ where $p, q \in \{1, -1\}$, four non intersecting boxes of size ℓ.



$$\eta_{\ell}^{(p,q)}(x) := \frac{1}{\ell^2} \sum_{y_1 = x_1}^{x_1 + \ell p e_1} \sum_{y_2 = x_2}^{x_2 + \ell q e_2} \eta(y_1, y_2) \tag{29}$$

particles density in $B^{\ell}_{pe_1,qe_2}(x)$. For fixed $u = (u_1, u_2) \in \mathbb{T}^2$ consider

$$i_{\varepsilon}^{p,q,u}(v) := \begin{cases} \frac{1}{\varepsilon^2} 1_{[u_1, u_1+\varepsilon) \times [u_2, u_2+\varepsilon)}(v_1, v_2) \text{ if } (p,q) = (1,1), \\ \frac{1}{\varepsilon^2} 1_{[u_1, u_1+\varepsilon) \times (u_2-\varepsilon, u_2]}(v_1, v_2) \text{ if } (p,q) = (1,-1), \\ \frac{1}{\varepsilon^2} 1_{(u_1-\varepsilon, u_1] \times [u_2, u_2+\varepsilon)}(v_1, v_2) \text{ if } (p,q) = (-1,+1), \\ \frac{1}{\varepsilon^2} 1_{(u_1-\varepsilon, u_1) \times (u_2-\varepsilon, u_2]}(v_1, v_2) \text{ if } (p,q) = (-1,-1). \end{cases} \tag{30}$$

Notation:

$$\pi_t(f) := \int_{\mathbb{T}^2} f(u) \rho_t(u) du, \ \pi_t^N(f) := \int_{\mathbb{T}^2} f(u)\, d\pi^N(\eta_t, du),$$

Proof: characterization of the limit points 3

$x \leftrightarrow \mathfrak{f}_x = \{x, x+e^{(1)}, x+e^{(1)}+e^{(2)}, x+e^{(2)}\}$, $f_x^{\circlearrowleft}$ and $f_x^{\circlearrowright}$. Note that

$$\eta^{(p,q)}_{s,\,\varepsilon N}(x) = \pi^N_s\left(i_\varepsilon^{(p,q,x/N)}\right) \tag{31}$$

Moreover since $\pi^N_t(f) \approx \pi_t(f)$ and from Lebesgue's differentiation theorem

$$\lim_{\varepsilon\to 0} \pi_s\left(i_\varepsilon^{(p,q,u)}\right) = \rho_s(u), \tag{32}$$

in the "macroscopic small ($\varepsilon \to 0$) but microscopically large $N >> 1$" box $B^{\varepsilon N}_{pe_1,qe_2}(x)$ we have that

$$\eta^{(p,q)}_{s,\,\varepsilon N}(x) \approx \rho(x/N, s). \tag{33}$$

Proof: characterization of the limit points 4

Then (28) follows from some bounds and proper algebraic manipulations that exploit the replacement lemma, i.e. for each $\delta > 0$,

$$\lim_{\varepsilon\to 0}\lim_{N\to\infty}\mathbb{P}_N\Bigg(\sup_{0\le t\le T}\Bigg|\int_0^t ds\sum_{\mathfrak{f}\in\mathcal{F}_N}\tau_{\mathfrak{f}}g(\eta_s)\sum_{(x,y)\in f^{\circlearrowleft}}G_N(x,y) \tag{34}$$

$$-\int_0^t ds\int_{\mathbb{T}^2}du\,\left(\alpha\pi_s^N\left(i_\varepsilon^{(-1,-1,u)}\right)\left(1-\pi_s^N\left(i_\varepsilon^{(1,-1,u)}\right)\right)\pi_s^N\left(i_\varepsilon^{(1,1,u)}\right)\left(1-\pi_s^N\left(i_\varepsilon^{(-1,1,u)}\right)\right)\right)\nabla^\perp\cdot G(u)$$

$$-\int_0^t ds\int_{\mathbb{T}^2}du\,\left(\alpha\left(1-\pi_s^N\left(i_\varepsilon^{(-1,-1,u)}\right)\right)\pi_s^N\left(i_\varepsilon^{(1,-1,u)}\right)\left(1-\pi_s^N\left(i_\varepsilon^{(1,1,u)}\right)\right)\pi_s^N\left(i_\varepsilon^{(-1,1,u)}\right)\right)\nabla^\perp\cdot G(u)\Bigg|>\delta\Bigg)=0$$

this allow to close the hydrodynamics of the current in terms of the empirical measure and replace $\eta_s(x)$ with $\mathbb{E}_{\rho(x/N,s)}(\eta(x))$.

<u>*Reminder*</u>: $g(\eta)=\alpha\eta(0)(1-\eta(e^{(1)}))\eta(e^{(1)}+e^{(2)})(1-\eta(e^{(2)}))+\alpha(1-\eta(0))\eta(e^{(1)})(1-\eta(e^{(1)}+e^{(2)}))\eta(e^{(2)})$ and $\sum_{\mathfrak{f}\in\mathcal{F}_N}\tau_{\mathfrak{f}}g(\eta_s)=\sum_{x\in V_N}\tau_x g(\eta_s)$.

Important comments

Of course, because of poor microscopic details, the derivation is a mathematical exercise more than a physical derivation, but the Hodge decomposition is not related to the stochastic nature:

- it comes from the geometrical nature of the microscopic current as discrete 1-form;
- Hodge decomposition is true both general for discrete vector field and continuous ones.

Therefore this microscopic structure could be a physical fact in the vortices formation.

Open questions

- experimental/numerical verification of the generalized Ficks law;
- develop of the large deviations theory for the joint current-density fluctuation;
- central limit theorem for the current;
- understanding if there exist or not reversible dynamics of this kind and studying minimal entropy production principle related to vortices;

On going work (if you are interested to join:)): mass transport in vortices

Consider transition rates perturbed by the presence of the external field

$$c^G_{x,y}(\eta) := c_{x,y}(\eta)e^{G_N(x,y)} \;\; G_N = O(1/N). \tag{35}$$

It will happen that

$$J^G(\rho) = -\nabla\rho - A(\rho)\nabla\rho + \sigma(\rho)G, \;\; \sigma(\rho) = 2\mathbb{E}_\rho(\eta(x)(1-\eta(y))), \tag{36}$$

regardless the intensity of the external field, vortices don't transport mass and verify Einstein's relations $D(\rho) = \sigma(\rho)f''(\rho) = \mathbb{I}$, $f(\rho)$ eq. free energy.

- it seems that outside the gradient condition the transport coefficients of a diffusive model are related only to a gradient part of current;
- we conjecture that asymmetric models obtained as strong asymmetric limit of (35) will still be diffusive.